Neurodivergent Learning & Sentiment Dashboard

- The Soloist | Stuti Mathur
- Goa Institute of Management | 2026
- Python-powered inclusive learning analytics | Mock prototype

The problem

- Generic learning dashboards may miss accessibility needs and declining engagement.
- Neurodivergent learners can benefit from individualized, voluntary support.
- Managers need understandable indicators without stigmatizing individuals.

Healthcare use case

- Illustrative L&D monitoring for hospitals, pharma and health-tech employers.
- Track cohort learning progress, sentiment and accommodation adequacy.
- Use attention signals to initiate supportive human check-ins—not personnel decisions.

Solution overview

- 800 × 600 desktop dashboard with four headline KPI cards.
- Learner progress bars; six sentiment dimensions; attention signals.
- Workforce outcomes and R-key refresh animation.

Illustrative dashboard KPIs

- 142 active learners | 73% average completion | 8.1/10 sentiment.
- 96% retention | 91% task completion | 34 Python projects.
- 38-minute focus sessions | 19 certifications.
- All figures are synthetic examples, not measured workplace outcomes.

Sentiment and alerts

- Engagement 82% | Confidence 76% | Belonging 70%.
- Stress 34% (lower is better) | Manager support 88% | Accommodation 65%.
- Example alerts: stalled progress, shorter sessions, low accommodation ratings.
- Alerts are illustrative, not automated clinical or HR diagnoses.

Technical architecture

- Synthetic Python constants → lists/tuples → drawing functions → Tkinter Canvas.
- Modular functions create cards, bars, text, alert panels and outcomes.
- R key briefly shows a refresh animation, then redraws unchanged values.

AI / ML disclosure

- No artificial intelligence or machine-learning model is used.
- No model training, inference, predictive scores or validation claims.
- Rule-style attention messages are fixed example data.
- Future AI extensions would require consent, fairness testing and evaluation.

Responsible deployment

- Fictional data only; avoid storing identifiable disability information.
- Require consent, accessibility, access control and aggregation in real deployments.
- Human review must remain central; no automated employment decisions.

Results, limitations and next steps

- Demonstrates Python modularity, visualisation and keyboard event handling.
- Not a live platform; no database, API, workplace pilot or impact evaluation.
- Next: user testing, secure ingestion, accessibility checks and governance.

Team and submission

- The Soloist — Stuti Mathur — Goa Institute of Management.
- Repository includes README, source, MIT license, architecture PDF and slides.
- Demo video must be recorded, uploaded unlisted and linked before submission.